

Subdivision cannot separate graph configuration homotopy types

Local formal research package for conjecture 00000004110

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Result. For every finite graph and every finite subdivision obtained by inserting vertices on edges, the continuous ordered and unordered configuration spaces are homeomorphic for every particle count. They are therefore homotopy equivalent for every count, in particular at $n = 2$. The source-bound refutation targets its additional assertion that the four-claw graph is the smallest example: no ordinary subdivision of that actual graph can be an example.

Original and scope

The source defines subdivision as inserting vertices on edges. It then asserts the existence of a graph and its proper subdivision whose configuration homologies are isomorphic for all n , but whose configuration spaces are not homotopy equivalent already for $n = 2$, with the four-claw graph as the smallest example. Both language versions express this same claim. The original does not explicitly restrict all graphs to finite ones. Our general finite-graph theorem therefore does not claim to settle an unrestricted infinite-graph existential by itself. Instead, the smallest-example assertion necessarily requires a separation witness whose original graph is the four-claw. We disprove that necessary witness clause. No additional homology computation or minimality search is needed to refute the whole conjunction. Arbitrary edgewise subdivisions of infinitely many original edges are outside the formal general theorem's scope.

The formal graph presentation consists of finite vertex and edge types and source/target maps. Vertices are discrete. Its realization is the quotient of the disjoint union of its vertices and one closed real interval $[0, 1]$ per edge, identifying the interval's endpoints with the designated vertices. This is the standard one-dimensional graph realization. Isolated vertices, loops and parallel edges are allowed. Restricting to simple graphs therefore remains covered. The result concerns continuous configurations on this realization, not selections of combinatorial vertices or a discretized cubical approximation.

A genuine insertion homeomorphism

For an edge e from u to v , insert a new vertex w and a second edge. The old edge label now ends at w , and the new edge runs from w to v . All other edges and incidences are preserved. On the realization, the collapse to the original graph sends the first edge's coordinate t to $t/2$, the second edge's coordinate t to $(1 + t)/2$, and the new vertex to the original edge's midpoint. Original vertices and unaffected edge coordinates remain fixed.

Its inverse sends an original edge point with $t \leq 1/2$ to the first edge at $2t$, and with $t > 1/2$ to the second edge at $2t - 1$. At the dividing point the two formulas agree in the quotient: the end of the first edge and the start of the second both equal w . The formal implementation uses globally defined clamped coordinates $\min(1, 2t)$ and $\max(0, 2t - 1)$ to establish continuity on the closed halves. The endpoint identifications are checked explicitly; quotient lifting yields continuous maps. Direct coordinate calculations, including the midpoint and both endpoints, prove both inverse identities. Thus the homeomorphism is constructed rather than assumed.

Presentation identity and finite subdivisions

Bijjective relabelings of vertices and edges that preserve endpoints induce a homeomorphism by applying the vertex bijection and carrying each interval, with unchanged coordinate, to its new edge label. The inverse uses the inverse bijections. Endpoint compatibility and continuity on each component are proved before quotient lifting.

An edge's orientation is a presentation choice for an undirected graph. Reversing selected edges swaps their source and target and replaces t by $1 - t$. This is continuous, preserves the actual endpoint relation and is an involution. The formal development further proves that an arbitrary bijection preserving each edge's unordered endpoint pair factors through such reorientation and an endpoint-preserving relabeling.

The inductive relation **Subdivision** $G \rightarrow H$ k is the finite closure generated by these actual presentation identities and one-edge insertions. Its index k counts edge insertions: identity, relabeling and orientation changes contribute zero; insertion contributes one; composition adds counts. A proper subdivision has $k > 0$. Its definition contains no homeomorphism or homotopy-equivalence hypothesis. Induction composes the already constructed elementary homeomorphisms. Any finite placement of new vertices in edge interiors gives precisely this combinatorial operation; the chosen midpoint parametrization is harmless because each resulting edge is a full closed interval with its endpoints glued.

Every ordinary four-claw subdivision

The actual four-claw presentation has vertex type $\text{Fin}(5)$, edge type $\text{Fin}(4)$, source constantly 0 and target $i + 1$ for edge i . Thus it has one center, four distinct leaves and exactly their four joining edges.

An ordinary subdivision replaces each edge by a nonempty finite path whose interior vertices are fresh and independent from the other replacement paths. This is the standard graph-theoretic convention: see Diestel, *Graph Theory*, sixth edition, Sections 1.3 and 1.7, in the author's Chapter 1 preview. The four-claw has only four edges, so its ordinary subdivisions always require finitely many insertions, irrespective of whether other graphs in the original existential are allowed to be infinite.

The formal path-replacement operation adds k interior vertices indexed by $\text{Fin}(k)$ and k fresh edges. The old selected edge becomes the first path edge. It joins the original source to interior vertex 0 when $k > 0$; fresh edge j joins interior vertex j to $j + 1$, or to the original target at the final step. For $k = 0$ the original edge stays intact. Induction on k proves this actual path presentation belongs to the insertion closure. At the induction step, insert one vertex on the first path edge; the explicit bijections shift every old interior index and fresh-edge index by one, and give the new vertex and new edge index zero. Every endpoint equation is checked.

Apply this replacement independently to all four original edges with arbitrary counts k_0, k_1, k_2, k_3 . The disjoint-sum label types make all added vertices and edges fresh. Ordinary subdivision membership then allows actual vertex and edge bijections preserving every unordered endpoint pair, hence arbitrary labels and orientations. Its structural witness consists of the four path lengths and these incidence bijections; it contains no homeomorphism, homotopy equivalence or insertion-closure premise. The proved coverage theorem derives the insertion closure, with count $k_0 + k_1 + k_2 + k_3$, from that ordinary finite-path structure. This covers the usual four independent finite-path replacement definition; independent semantic review must verify this source correspondence.

Actual configuration spaces

For a topological space X , define

$$F_n(X) = \{f : \{0, \dots, n-1\} \longrightarrow X \mid f \text{ is injective}\},$$

with the subspace topology inherited from the finite product X^n . The unordered space $C_n(X)$ is the orbit quotient under permutations of the indices, with its quotient topology. The formal orbit relation is the actual existence of a permutation taking one configuration to the other.

If $h : X \rightarrow Y$ is a homeomorphism, the coordinatewise map $f \mapsto h \circ f$ preserves injectivity. Its inverse is $g \mapsto h^{-1} \circ g$; both are continuous in the product/subspace topology. It commutes with every index permutation, hence descends to a homeomorphism of the unordered orbit quotients, again with a checked continuous inverse. This construction works uniformly for every $n \in \mathbb{N}$, including zero.

Theorem 1. *For finite graph presentations G, H and any finite subdivision H of G , $F_n(|G|)$ and $F_n(|H|)$ are homeomorphic, as are $C_n(|G|)$ and $C_n(|H|)$, for every $n \in \mathbb{N}$.*

Proof. Compose the insertion and presentation homeomorphisms and apply the coordinatewise and orbit-quotient constructions above. $\square$

Mathlib’s `ContinuousMap.HomotopyEquiv` consists of continuous maps in both directions whose inverse composites are homotopic to the identities. Its `Homeomorph.toHomotopyEquiv` supplies these witnesses from a homeomorphism: the composites equal the identities, so constant interval homotopies suffice. Applying it to the theorem’s constructed homeomorphisms proves the actual homotopy equivalences. Therefore the negated existential separation statements at $n = 2$ follow immediately for both conventions.

Lean correspondence and verification

1. `ConnectedPartial.lean`: actual ordered and unordered configurations, endpoint-cell realization, explicit insertion and inverse continuity.
2. `Relabeling.lean`: vertex/edge relabeling homeomorphism.
3. `Reorientation.lean`: genuine interval reversal and graph-cell identity.
4. `Subdivision.lean`: finite graph presentations, actual undirected presentation identity, counted insertion closure and configuration homeomorphisms.
5. `Disproof.lean`: actual Mathlib homotopy equivalences and both finite-graph separation contradictions.
6. `PathReplacement.lean`: concrete finite paths and their proved membership in the insertion closure.
7. `FourClaw.lean`: actual $K_{1,4}$, ordinary four-path subdivision coverage and both ordered and unordered four-claw separation contradictions.

The package pins Lean 4.33.0 and every public dependency by Git commit. The graph and configuration bridges are proved; no central bridge is an extra assumption. Independent semantic review and the manager’s deterministic build, type, axiom, PDF and fresh-environment checks are required before local acceptance. This report itself makes no claim of external submission or accepted solve status.